

Here's a comprehensive overview of **Session 9: Storage Area Network (SAN)** theory, followed by **40 MCQs** (10 Easy, 20 Medium, 10 Hard) with four options, correct answers, and explanations.

☒ Theory: Storage Area Network (SAN)

1. Configuring a SAN (using FreeNAS/TrueNAS)

- FreeNAS (TrueNAS) can be set up as a SAN via iSCSI: create portal ☒ initiators ☒ targets ☒ extents ☒ associate extent to target ([reddit.com](https://www.reddit.com/r/truenas), [truenas.com](https://www.truenas.com)).
- Use ZFS zvols (block devices) to define extents; FreeNAS allows quotas, thin provisioning, and advanced options like dataset compression ([ixsystems.com](https://www.ixsystems.com)).

2. Using SAN for High Availability

- High-availability (HA) in TrueNAS is achieved using dual controllers with CARP or RSF-1 clustering and shared storage (e.g., clusters across iSCSI via ZFS) ([high-availability.com](https://www.high-availability.com)).
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ZFS HA uses multipathing and dual-head SAN design enabling failover without downtime ([reddit.com](#)).

3. ZFS Volume Configuration

- Configure zvols via GUI: set size (with soft/hard quota), enable compression (lz4), launch zvol as block device for iSCSI ([ixsystems.com](#)).
- Import existing ZFS pools for reuse upon reinstallation ([ixsystems.com](#)).

4. IP-Based Storage Communication

- iSCSI uses TCP/IP (over Ethernet, often 10 GbE) to transport SCSI commands. Configure network ports, VLANs, MTU, and performance tuning.

5. Object Storage Services

- Offers scalable, metadata-rich storage via HTTP APIs. Protocols like S3, Swift. Common for backups, big data, cloud archives.

☒ Easy (10 MCQs)

1.

****Which service sets up a SAN in FreeNAS?****

- A) SMB
- B) NFS
- C) iSCSI
- D) FTP

Answer: C

Explanation: SAN uses iSCSI for block-level storage (ixsystems.com, hostman.com, hardforum.com).

2.

****ZFS blocks used for SAN extents are called:****

- A) Datasets
- B) Snapshots
- C) zvols
- D) Pools

Answer: C

Explanation: zvol is a volume for block storage (ixsystems.com).

3.

To create a zvol you:

- A) Use Volume Manager
- B) Create a snapshot
- C) Launch FreeBSD shell
- D) Download templates

Answer: A

Explanation: Use Volume Manager in GUI (en.wikipedia.org, ixsystems.com).

4.

****HA in TrueNAS is implemented using:****

- A) Single server
- B) Dual controllers & CARP/RSF-1
- C) FreeBSD jail
- D) VLAN assignation

Answer: B

Explanation: HA uses clustering with CARP or RSF-1 (high-availability.com, truenas.com).

5.

****iSCSI transports SCSI commands over:****

- A) Fiber channel only
- B) TCP/IP
- C) USB
- D) Infiniband

Answer: B

6.

Thin provisioning in zvol is enabled by:

- A) Sparseness
- B) Compression
- C) Snapshot
- D) Multipathing

Answer: A

Explanation: Sparse option provides thin provisioning (ixsystems.com).

7.

****Compression level recommended is:****

- A) Off
- B) lz4
- C) gzip-9
- D) zip

Answer: B

Explanation: lz4 is efficient with low impact (en.wikipedia.org, ixsystems.com).

8.

****High availability ZFS uses:****

- A) Single-head pool
- B) Dual-head pool
- C) Mirror only
- D) RAID-Z

Answer: B

Explanation: Dual-head with multipathing (reddit.com, ixsystems.com, forum.proxmox.com).

9.

Object storage uses:

- A) iSCSI
- B) S3/Swift APIs
- C) NFS
- D) CIFS

Answer: B

10.

IP-based SAN commonly uses:

- A) 1 Gb Ethernet only
- B) 10 Gb Ethernet

- C) SATA cables
- D) HDMI cables

Answer: B

☒ Medium (20 MCQs)

11.

****SAN portal ☒ initiators ☒ targets ☒ extents sequence is for:****

- A) SMB share
- B) iSCSI setup
- C) Snapshot creation
- D) ZFS scraping

Answer: B

Explanation: Standard iSCSI configuration (johnkeen.tech, ixsystems.com).

12.

****Before creating extent, one must:****

- A) Create snapshot
- B) Create zvol
- C) Export pool
- D) Disable compression

Answer: B

Explanation: zvol needed for extent (en.wikipedia.org, ixsystems.com).

13.

****To reuse disks after FreeNAS reinstall, use:****

- A) zfs import
- B) format
- C) delete pool
- D) clone

Answer: A

Explanation: Import existing pool via GUI or CLI (ixsystems.com).

14.

ZFS quotas prevent:

- A) Unlimited writes elsewhere
- B) Exceeding pool usage
- C) File fragmentation
- D) Network leaks

Answer: B

15.

TrueNAS HA uses VIP for:

- A) Storage migration
- B) iSCSI traffic only
- C) VLAN trunking
- D) CPU scaling

Answer: A

Explanation: VIP migrates with pool on failover (hardforum.com, high-availability.com, high-availability.com).

16.

ZFS multipathing improves:

- A) Network speed
- B) I/O resilience
- C) Snapshot frequency

D) CPU usage

Answer: B

17.

ZFS dataset vs zvol difference:

A) Dataset = file system; zvol = block device

B) Opposite

C) Same

D) zvol for snapshots only

Answer: A

18.

FreeNAS backups use:

A) Samba

B) Rsync/replication tasks

C) USB drives only

D) Manual copy

Answer: B

Explanation: Periodic snapshot and replication via Rsync (en.wikipedia.org, ixsystems.com).

19.

****Object storage suited for:****

A) OS installation

B) Unstructured data at scale

C) iSCSI block ops

D) VM disks

Answer: B

20.

zvol block size affects:

- A) Color scheme
- B) Performance alignment
- C) Compression disabled
- D) Network speed

Answer: B

Explanation: Choose blocksize matching workload (ixsystems.com, serverwatch.com).

21.

****iSCSI target's NAA IDs synced for:****

- A) VIP assign
- B) HA failover
- C) Pool encryption
- D) NAS only

Answer: B

Explanation: Ensures same identifier across cluster nodes (high-availability.com).

22.

CARP enables HA by:

- A) CPU resume
- B) Shared VIP
- C) Disk leadership
- D) NFS sync

Answer: B

23.

TrueNAS HA alternative to CARP:

- A) Samba
- B) RSF-1
- C) Zvol

D) CIFS

Answer: B

24.

Replication tasks enforce:

A) GUI usage

B) Offsite sync

C) Pool creation

D) CLI logging

Answer: B

25.

iSCSI Jain vs zvol: termination:

A) VLAN

B) Block device

C) Snapshot

D) Encryption

Answer: B

26.

Multipathing requires:

A) Bridge

B) Multipath config

C) Backup cables

D) USB passthrough

Answer: B

27.

Snapshot cloning creates:

- A) Linked copy
- B) Full backup
- C) zvol only
- D) Object store

Answer: A

28.

ZFS copy-on-write ensures:

- A) Instant I/O
- B) Data integrity
- C) CPU offload
- D) x86 only

Answer: B

29.

Object storage scalability is:

- A) Limited to TB
- B) Effectively unlimited
- C) Only for SAN
- D) CPU-limited

Answer: B

30.

iSCSI initiator runs on:

- A) FreeNAS
- B) Client host OS
- C) Switch
- D) HBA only

Answer: B

☒ Hard (10 MCQs)

31.

****To configure SAN in FreeNAS, final step is:****

- A) Reboot
- B) Associate extent with target
- C) Disable GUI
- D) Snapshot

Answer: B

Explanation: Last step in iSCSI setup (high-availability.com, truenas.com, high-availability.com, ixsystems.com, en.wikipedia.org, truenas.com).

32.

****High-availability SAN fails over via:****

- A) Manual reboot
- B) VIP migration
- C) Snapshots
- D) Pool deletion

Answer: B

Explanation: VIP moves to active node (high-availability.com).

33.

RSF-1 Active-Active clustering means:*

- A) Only one pool active
- B) All pools available on both

- C) No failover
- D) Container only

Answer: B

*From clustering docs (hardforum.com).

34.

ZFS pool max size is:

- A) 16 EB
- B) 1 PB
- C) 1 TB
- D) 16 GB

Answer: A

ZFS massive scalability .

35.

Blocksize mismatch causes:

- A) Snapshot fail
- B) I/O inefficiency
- C) Encryption fail
- D) No effect

Answer: B

36.

DRBD vs iSCSI HA: DRBD is:

- A) NFS
- B) Block-level replication
- C) NAS protocol
- D) GUI

Answer: B

37.

TrueNAS cluster heartbeats go over:

- A) Internet
- B) Network & disk
- C) USB
- D) Serial only

Answer: B

Supports cluster HA via disk & network heartbeats (en.wikipedia.org, hostman.com, high-availability.com, high-availability.com, serverwatch.com).

38.

****FreeNAS multipath not HA because:****

- A) Single controller
- B) Shared storage not clustered
- C) zvol-only
- D) No VIP

Answer: B

39.

Object storage uses HTTP method:

- A) PUT/GET
- B) SMB
- C) iSCSI
- D) NFS

Answer: A

40.

FreeNAS sparse zvol warns:

- A) Uptime limit

- B) Over 80% use
- C) Snapshot disabled
- D) HA disabled

Answer: B

zvol not created if >80% unless forced (ixsystems.com, forum.proxmox.com, high-availability.com, high-availability.com).

Let me know if you'd like deep explanations for any question or are ready to move on to Session 10!